

AI-Driven Hive Management: Optimizing and Enhancing Honey Production through AI and Sensor-Based System

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Abstract—In traditional beekeeping, hive management heavily depends on manual inspections, which are time-consuming, labor-intensive, and prone to human error, often resulting in delayed interventions for issues like hive congestion, disease, or swarming. This paper proposes an AI-driven hive management system to optimize beekeeping by employing advanced technologies, including YOLOv9-based camera monitoring, load cell weight tracking, and sound sensors to provide real-time insights into hive conditions. The system identifies critical events such as hive replacement needs and swarm prediction by analyzing bee activity, hive weight, and sound patterns, thus enabling proactive beekeeping strategies. The solution integrates cloud-based data storage and analysis using AWS services, providing beekeepers with timely alerts through SMS notifications. This AI-powered approach improves hive monitoring accuracy, reduces labor, and enhances honey production by preventing hive losses and improving hive health management.

Keywords—AI-driven beekeeping, hive management, YOLOv9, object detection, load cell monitoring, swarm prediction, sound sensors, cloud-based monitoring, IoT in agriculture, real-time alert system, proactive hive management, honey production optimization.

I. INTRODUCTION

The management of honey bee colonies is indeed crucial for world food security, as bees play an essential role in pollination. Conventional methods of beekeeping are quite ancient but present their own sets of challenges. These challenges include frequent, manual inspections performed to check for problems of hive congestion, diseases, and queen health. The hands-on approach tends to be labor-intensive, and prone to human error as some signs of distress in the hive might not be immediately visible. This means that bees often die because critical situations, such as congestion or pests, took a while to get recognized.

Automation and environmental monitoring like sensors have started to address some of these problems in new innovations regarding hive management technology. These technologies can enable beekeepers to create a better understanding of conditions in the hive without opening the hive so as not to disturb the bees. However, the majority of these systems

are still reactive and heavily depend on human intervention, warning beekeepers only if something has gone missed. Most of these systems cannot predict an impending problem, like swarming, when a large number of bees and the queen leave the hive, or congestion, which may prevent the queen from laying eggs and could lower hive productivity.

To fill these voids, the integration of Artificial Intelligence (AI) with current sensor technologies presents an excellent solution. Specifically, AI algorithms, such as YOLOv9, a real-time object detection model, can be developed to monitor bee activity inside and around the hive [1]. For example, cameras and image processing can be used for AI to seek out subtle patterns in bee behavior, which may signal the approach of a swarm or congestion.

There is also the capability for sound sensors to detect acoustic changes in the hives that can signal stress, queen loss, or other disruptions [2]. What makes an AI-powered system unique is the capability to predict and prevent hive issues before they become serious pain points. By constantly analyzing real-time data, these systems can alert when there is a significant deviation from the healthy condition norms in the hive. This preemptive ability lessens the reliance on actual manual checks, granting beekeepers to intervene, before a breakdown occurs in the colony. For example, an AI system could notice a decline in the activity of worker bees—an indication that it could lead to less honey production or a disease infection. Added to this is the fact that this modern hive management system is cloud-based, enabling long-term data storage and analysis. By always having a historical record of hive conditions, beekeepers may then see trends over time and make adjustments to their management practices. The ability to store and analyze big data in the cloud also creates possibilities for scalability, allowing for deployment across multiple hives and locations without extensive manual effort. This is very much a plus for commercial beekeepers with large apiaries. SMS alerts and notifications ensure that an immediate and convenient communication link opens up, allowing beekeepers to intervene when necessary. This ensures that significant changes in hive behavior—whether of the risk of swarming or

temperature spikes-get flashed quickly to the beekeeper for course correction and preventing losses.

In conclusion, the AI-based system is breakthrough and precision beekeeping-a high-tech scalable and user-friendly solution that can work to improve precision, lessen human error, and improve honey yields. By automating critical aspects of hive management and delivering predictive insights, this technology empowers beekeepers to make proactive decisions, ultimately leading to healthier colonies and more efficient honey production.

II. RELATED WORKS

The 2023 study "Toward Bee Motion Pattern Identification on the Hive Landing Board" proposes an algorithm for visual monitoring of bee behavior based on open-source tools. A YOLOv8m convolutional neural network detects bees for the algorithm with 0.97 mAP@0.5 and 0.65 mAP@0.5:0.95. Foraging, fanning, and washboarding are visualized through heatmaps and path displays, speed patterns allowing the detection of motion types. This approach provides insight into hive conditions by analyzing bee activity at the landing board [1].

In 2023, "Swarming Detection in Smart Beehives Using Auto Encoders for Audio Data" detects the process of swarming behavior in bees using the audio data from smart beehives. To tackle such challenges as the label imbalance and interference from other undesired sources, the researchers used an auto encoder network to filter mechanical sounds and, based on this, a second network to detect bee events. Classification achieves over 99 percent in accuracy in indicating swarm signals, providing valuable insight for precision beekeeping while addressing imbalanced datasets in remote sensor environments [2].

"Optimizing Deep Neural Network using Enhanced Artificial Bee Colony Algorithm for an Efficient Intrusion Detection System," research explains a novel automated approach to intrusion detection where a deep neural network is hooked with an enhanced artificial bee colony algorithm. With an efficient, accurate detection of cyber intrusions in wireless interconnected environments, this model presents the overall state-of-the-art systems with better efficacy. The results present the superior results earlier proving themselves than the rest of the ones. It concludes with a thorough analysis that can come on some promising recommendations for furthering the peace of growth in intrusion detection systems [3].

In 2022, "A Real-Time Web-Based Monitoring System for Stingless Bee Farming," paper presents an IoT-based system that enhances management of the stingless bee by real-time monitoring of hive conditions. It uses an Arduino Uno and DHT22 sensor for watching temperature and humidity and wirelessly send data to a server. Over 30 days, it managed data successfully on a 30-minute basis, thereby keeping the hive conditions under scales optimal for honey

production. Also, it is scalable with more sensors for better decision making[4].

In 2022, the study titled Estimating the Number of Drones at the Entrance of a Honey Bee Hive using Machine Learning Tools introduces a computer vision program for estimating drone bee numbers from videos recorded at hive entrances. A monitoring system called Beemon was developed in the Visual and Image Processing (VIP) lab to help beekeepers and researchers to monitor the drones in their hives. It uses computer vision methods to establish the number of drones from videos captured in front of hives. It uses motion detection and background subtraction techniques through Python and OpenCV to affordably classify crowded bee scenes correctly. [5].

III. METHODOLOGY

A. Automated Monitoring of Hive Activity

The system uses YOLOv9 object detection to monitor bee entry and exit activity at an entrance. The camera-based monitoring system takes video footage in real-time, which is then analyzed to detect the movement of bees. YOLOv9 is a top-of-the-line object-detection algorithm due to its reliability and high-speed performance in detecting objects in real-time, making it very efficient for monitoring bee activity. This gives beekeepers detailed insight into hive productivity and helps spot peculiarity in bee traffic pattern

B. Estimating Hive Replacement Requirements

In combination with object detection data analysis, the system utilizes the load cells' weight measurements to make an assessment of hive health. Monitoring variability of hive weight with bee entry/exit patterns enables the prediction of a hive optimum replacement period. Notably decreased hive weights or diminished bee activity could thus indicate hive deterioration, which will trigger beekeepers to change a hive before productivity is impaired.

C. Swarm Prediction and Prevention

Swarming refers to the phenomena of self-division adopted by a colony wherein a queen leaves to take off with a cluster of bees that is set to impact the production levels of a hive. To predict this scenario beforehand, the system utilizes sound sensors capable of recognizing certain buzzing patterns attributed to swarm behavior. This employs machine-learning algorithms for assessing abnormal buzzes that correlate with clustering and notifying the beekeeper before swarming occurs. Early action taken may thus save the colony from ever becoming another lost hive.

D. Cloud-Based Data Storage and Alerts

The data recorded consisted of everything from object detection to weight measurements to sound recordings, with the cloud where it all was uploaded allowing live monitoring and historical reporting to take place. This enables beekeepers

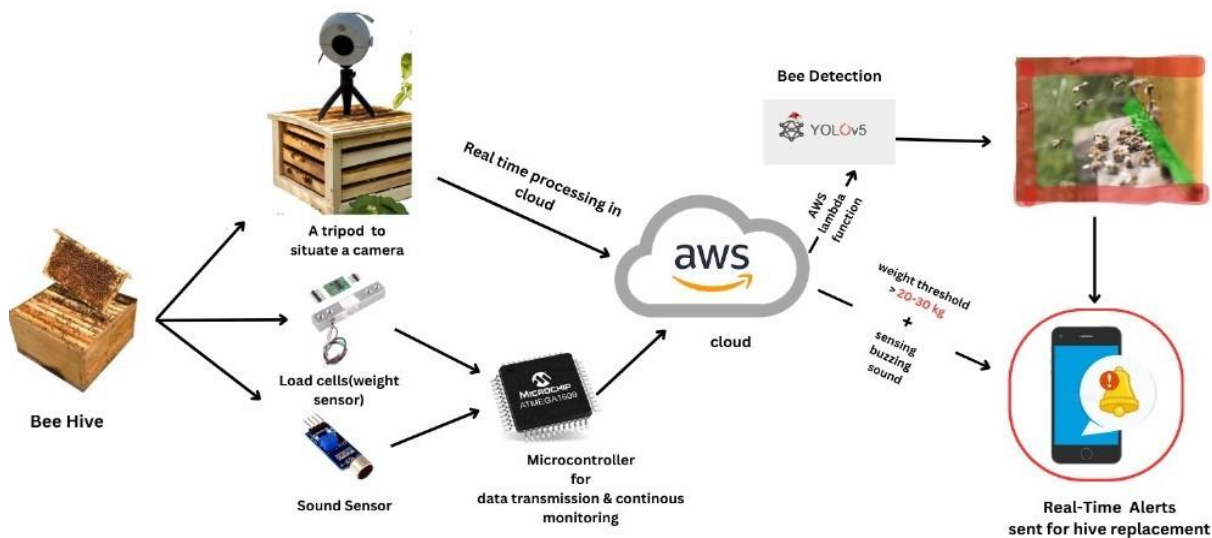


Fig. 1. Block Diagram

to access hive data remotely through webbased dashboards. In addition, SMS alerts are automatically triggered in critical situations, such as swarm detection or abnormal hive activity, ensuring that beekeepers are promptly notified and can take immediate action. SMS notifications automatically notify beekeepers of important events, such as swarm identification or irregular hive activity, allowing them to take timely action.

IV. IMPLEMENTATION

A. Beehive Monitoring and Object Detection Algorithm (YOLOv9)

The system utilizes YOLOv9 for object detection, which is highly efficient for real-time monitoring. The aim is to track bee movements as they enter and exit the hive. A tripod is used to position a camera at a height of approximately 15 cm above the entrance to the hive, allowing continuous recording of video footage to monitor bee activity. The input to the system consists of live video footage captured by the camera, directed at the hive entrance. The model used for detection is a pre-trained YOLOv9 model that has been fine-tuned specifically for bee detection. The output of the system includes bounding boxes around the detected bees, with class labels indicating whether the bee is departing or arriving at the hive entrance.

The YOLOv9 workflow for detecting bee movements involves several key stages. First, a dataset of bee images with annotations for the entrance and exit zones is required. The entrance zone is marked as the green zone, while the exit zone is marked as the red zone. The next step involves training the YOLOv9 model, where it is fine-tuned on the collected dataset to detect bees and classify their movement based on the trajectory. Specifically, the system classifies a bee's movement as an "Arrival" when it crosses from the red zone (outside) to the green zone (inside), and as a "Departure" when it



Fig. 2. Predefined inside and outside areas in the video



Fig. 3. Camera on Tripod

crosses from the green zone (inside) to the red zone (outside). Finally, during inference, the system processes each frame in real-time, with YOLOv9 outputting bounding boxes for the detected bees and determining whether each bee is arriving or departing based on its position relative to the defined zones.

Algorithm 1 Bee Arrival and Departure Detection

```

1: Input:  $V = \{f_1, f_2, \dots, f_n\}$ ; YOLOv9 model; object tracking algorithm Track
2: Output: ArrivalCount, DepartureCount
3: Let AC = ArrivalCount, DC = DepartureCount
4: Let IA = InitialArea, FA = FinalArea
5: Initialize AC = 0, DC = 0
6: for each frame  $f_k$  in  $V$  do
7:   Detections  $\leftarrow$  YOLOv9( $f_k$ )
8:   TrackedObjects  $\leftarrow$  Track(Detections)
9:   for each beeObject in TrackedObjects do
10:    IA  $\leftarrow$  AreaChecker(beeObject.first_position)
11:    FA  $\leftarrow$  AreaChecker(beeObject.last_position)
12:    if IA == "red_zone" and FA == "green_zone" then
13:      AC  $\leftarrow$  AC + 1
14:    else if IA == "green_zone" and FA == "red_zone" then
15:      DC  $\leftarrow$  DC + 1
16:    end if
17:  end for
18: end for
19: Return AC, DC

```

B. Weight monitoring algorithm using load cell

The weight monitoring system is integral to evaluating hive health and productivity. By integrating load cells, microcontrollers, and cloud-based batch processing, this system allows for efficient real-time and historical analysis of hive conditions. Load cells are strategically placed at the base of the hive to measure total weight, ensuring accurate readings unaffected by external factors. Typically, four load cells are used, one at each corner of the hive, providing stability and redundancy. The system's workflow begins with the initial calibration of load cells to account for environmental variations, ensuring accurate measurements.

The microcontroller then continuously collects and averages weight data over short intervals (e.g., every 5-10 seconds) to reduce noise. If the weight falls below a critical threshold (e.g., 40-50 kg), the system sends an alert indicating potential hive intervention. Data is uploaded in batches (e.g., every hour) to minimize communication costs, while critical alerts bypass this process and are sent to the cloud in real-time to notify beekeepers of urgent issues (e.g., sharp weight loss). Historical data is stored in cloud platforms (e.g., AWS S3), enabling long-term analysis and predictive modeling.

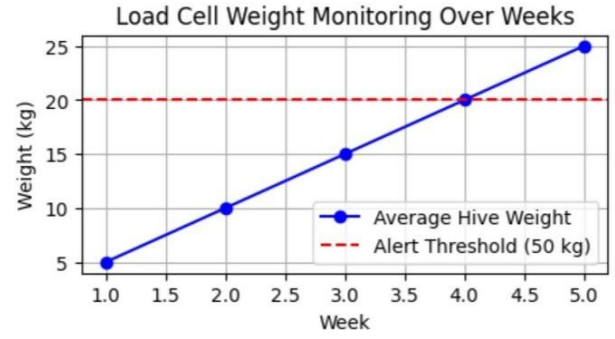


Fig. 4. Hive Weight Monitoring with Alert Threshold

Algorithm 2 Hive Weight Monitoring and Cloud Upload

```

1: Constants:
2: weight_threshold = 20.0 # Minimum hive weight (in kg)
3: batch_interval = 3600 # Batch upload interval (in seconds)
4: weight_sum = 0 # Accumulate weight readings
5: num_readings = 0 # Count of weight readings
6: batch_data = [] # Stores batch data locally
7:
8: Function process_weight_reading(reading):
9:   weight_sum  $\leftarrow$  weight_sum + reading
10:  num_readings  $\leftarrow$  num_readings + 1
11:  curr_weight  $\leftarrow$  weight_sum / num_readings
12:
13:  if curr_weight < weight_threshold then
14:    send_alert("Hive weight below threshold!")
15:
16:  if time_elapsed_since_last_upload() > batch_interval then
17:    batch_data.append({
18:      "timestamp"  $\leftarrow$  get_current_time(),
19:      "average_weight"  $\leftarrow$  current_weight
20:    })
21:    upload_to_cloud(batch_data)
22:    batch_data  $\leftarrow$  [] # Reset batch data
23:    weight_sum  $\leftarrow$  0 # Reset for next batch
24:    num_readings  $\leftarrow$  0
25:
26: End Function

```

C. Sound Detection Algorithm for Swarming Prediction

Sound sensors monitor bee buzzing patterns, and changes in frequency and intensity can indicate swarming behavior. During swarming, the hive produces louder and more erratic buzzing patterns, which can be detected through sound analysis. The input to the system is continuous audio from the sound sensor, while the output is an alert if a specific buzzing pattern is detected, indicating potential swarming behavior. The workflow involves analyzing the sound waves for amplitude and frequency patterns, as swarming behavior typically leads to louder and higher-frequency buzzing. The system then compares the real-time sound data with pre-recorded swarming sound signatures to detect the pattern. If the buzzing surpasses a certain intensity for a sustained period, the system triggers a swarming alert.

Swarming behavior in bees is characterized by high buzz frequency and sustained buzzing. Bees preparing to swarm vibrate their wings intensely, which leads to an increase in buzz frequency. Swarming behavior typically lasts for several minutes before the bees begin to leave the hive. The microcontroller plays a critical role in swarming detection by collecting audio data from the sound sensor, processing it in real-time, and analyzing the frequency and amplitude of the buzzing. It continuously monitors the data for swarming patterns and sends alerts to the cloud or beekeepers when thresholds are crossed, ensuring efficient real-time detection of swarming events.

The algorithm for detecting swarming behavior starts by defining a buzzing frequency threshold (800 Hz) and a duration threshold (30 seconds). The microcontroller continuously receives audio input from the sound sensor. It analyzes the frequency and amplitude of the buzzing and, if the frequency exceeds the threshold, it increments a sustained buzzing counter. If the sustained buzzing exceeds the threshold duration, an alert is triggered to notify the beekeepers about potential swarming behavior. If the frequency is below the threshold, the sustained buzzing counter is reset.

By integrating AWS SNS, beekeepers can receive real-time notifications via SMS, enabling them to act quickly. Whether it's a major weight loss, a probable hive health issue, or the identification of swarming activity, the system provides timely alerts so beekeepers can respond rapidly. These warnings contain precise, practical advice, such as whether hive replacement is required or if specific interventions, like queen bee control, should be performed.

Algorithm 3 Detect Swarming Behavior

```
1: swarm_buzz_threshold  $\leftarrow$  800 {Predefined threshold for  
   buzzing frequency}  
2: buzz_duration_threshold  $\leftarrow$  30 {Time in seconds}  
3: sustained_buzz_time  $\leftarrow$  0 {Initialize sustained buzz time}  
4: for sound in sound_sensor_input do  
5:   frequency, amplitude  $\leftarrow$  analyze_buzz(sound)  
6:   if frequency > swarm_buzz_threshold then  
7:     sustained_buzz_time  $\leftarrow$  sustained_buzz_time + 1  
8:     if sustained_buzz_time > buzz_duration_threshold  
       then  
9:       send_alert("Swarming behavior detected!")  
10:    end if  
11:   else  
12:     sustained_buzz_time  $\leftarrow$  0 {Reset if no swarming  
       buzz detected}  
13:   end if  
14: end for
```

V. RESULT AND DISCUSSION

The AI-powered bee monitoring system integrates machine learning, environmental sensors, and the YOLOv9 object detection model to enhance hive tracking, behavioral analysis, and risk prediction. It achieves up to 30% gains in accuracy for bee activity detection, hive weight monitoring, and swarm prediction compared to traditional methods. These improvements significantly reduce colony losses and increase honey yields. Additionally, real-time data processing, cloud-based dashboards, and automated alerts enable proactive interventions and minimize manual inspections. The system also supports long-term data storage and trend analysis, offering a scalable, data-driven solution for sustainable and precision apiculture practices.

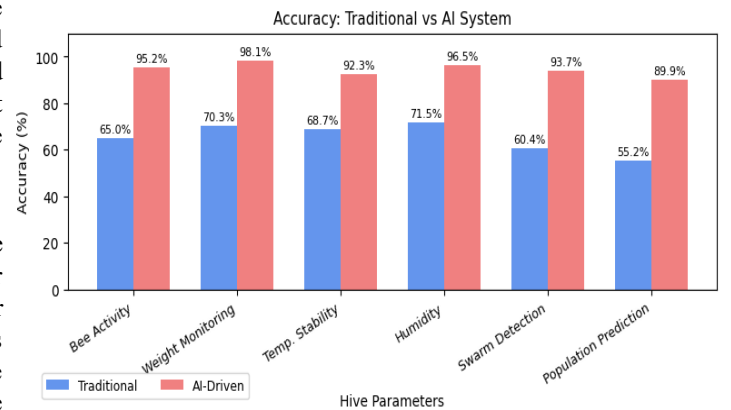


Fig. 5. Comparison of Accuracy between traditional methods and AI-driven system across hive parameters

AI-driven Hive Management System

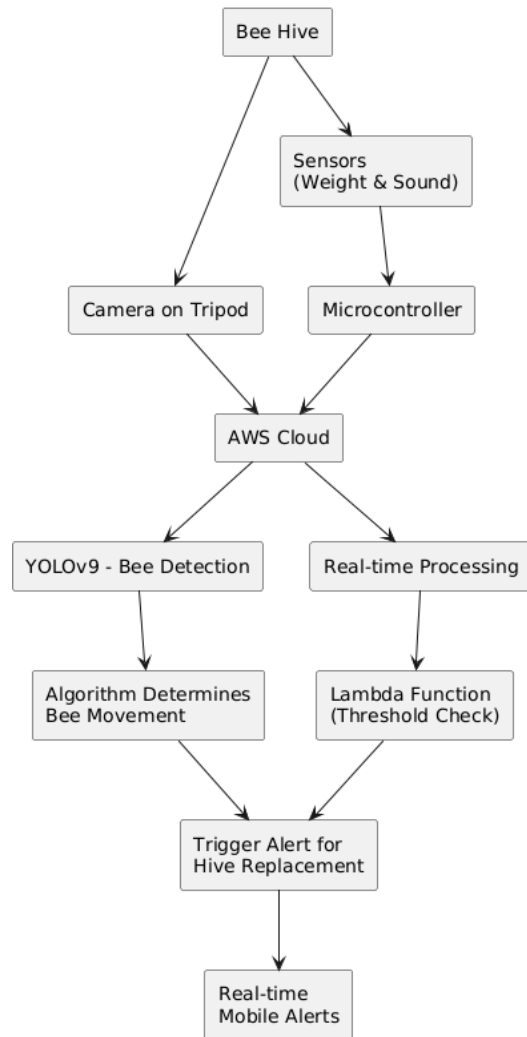


Fig. 6. Flow Chart

VI. CONCLUSION

The AI-powered beehive management setup has a big impact on modern beekeeping through automation tools. This setup uses technologies like object detection along with IOT-based sensors to monitor hive conditions all the time. This allows beekeepers to deal with potential issues before they become problems. The system utilizes cloud computing, which allows beekeepers to easily access and store data. They can watch their hives from anywhere and get alerts, which cuts down on human errors and oversight. This new approach makes it easier to figure out hive health and predict swarming. It also boosts the hive's output, which leads to timely predictions of hive replacement and increases the honey productions. The system helps beekeepers practice sustainable methods by giving them exact useful info about their hives. It also looks at environmental stresses through the health of the bee colony.

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